

The Utility Problem

With the usual challenges of electricity distribution operations along with the introduction of distributed energy, building electrification and transportation electrification, utilities are required to change their organizations to address both old and new needs. These factors increase the complexity of managing the grid as well as the demand for electricity, which are both vital to address. In order to address these challenges, a data-driven approach is required, which is possible as the utility has access to GIS, connectivity, and time-series data, among many other data sources. However, this data only becomes valuable when it is analyzed and/or visualized in a productive manner. Visualization of the data can enable the utilities to gain significant value in asset management, vegetation management, EV infrastructure planning, plus many more important use cases.

Despite utilities having access to all the required data, there is still the obstacle of going from the raw data to valuable data visualization. Utility source data is spread out in disparate source systems, lacks synchronization for time-series and GIS/connectivity data, and is riddled with errors. All three of the complex problems must be addressed to produce a final product that provides value to the utility, but unfortunately, this process often takes an extended period of time, is costly and requires significant labour resources without the right tools.

The Data Solutions and Methods

As mentioned above, the data must be brought together from disparate source systems, synchronized and cleansed to provide valuable insights. The Awesense Platform is capable of performing all of these operations at an accelerated rate and at a lower cost than industry standards.

Disparate source systems for different datasets are often a significant hurdle to data analysis within a utility. The Awesense Platform eradicates this challenge, bringing together the siloed datasets into a common framework in our Energy Data Model (EDM). In order to visualize the data productively, the GIS and time series data must be synchronized and the time series data must be synchronized with itself (fixing offsets and time intervals in the data model). This is all done through Awesense's data engine, in which the time series data is aligned and GIS and time series are synchronized to enable situational awareness of the full electricity distribution grid service area. Finally, inaccurate data must be cleansed in order to provide any value from the insights gained from the visualizations. This process is again completed within the Awesense data engine: validating, estimating and correcting various types of error within the data. Incorrect or inaccurate data will produce incorrect insights and therefore could be more detrimental to the company's decisions.

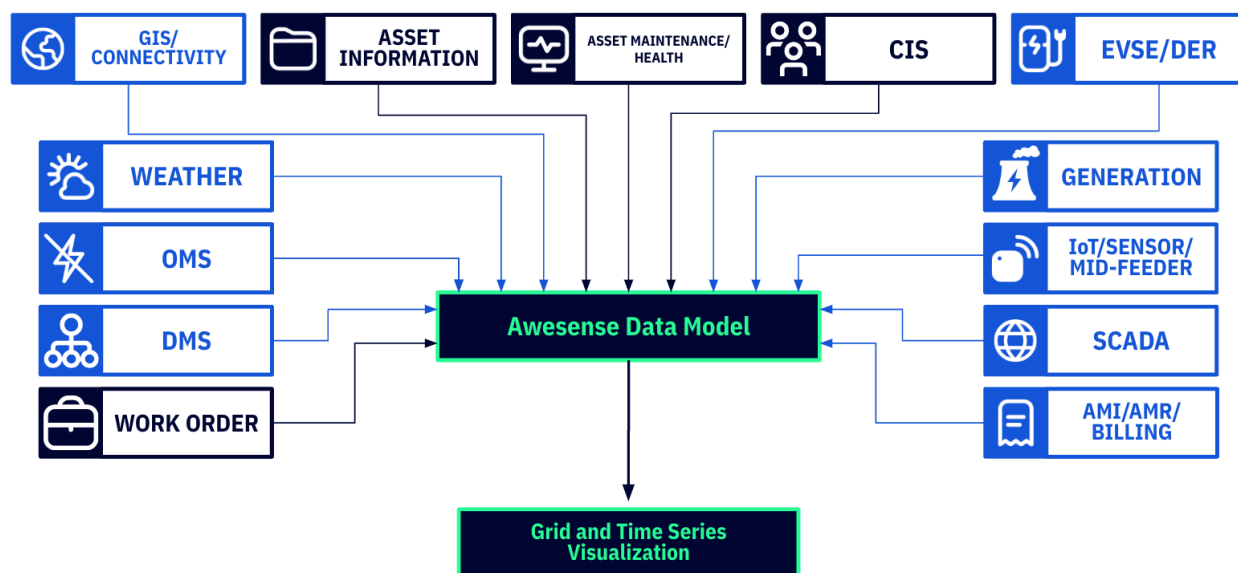
It is important to note that once these three processes have been completed, the data is much more powerful and can be utilized in a breadth of different visualizations. In Awesense's Energy Transition Platform, the data is overlaid on a map and time-series data line graphs can be produced by clicking on each component. This enables easy and efficient awareness of the

current state of the electrical grid along with contextual data browsing. With this capability, previously unknown relations and implications can be revealed such as change in power flow based on outside temperature or a switch in energy direction based on grid switch state. In addition, this format of the data can be used in a wide variety of ways after it has been ingested into the Awesense Energy Data Model (EDM).

Awesense's front-end application, True Grid Intelligence (TGI), exhibits many pre-built visualizations and allows for many custom visualizations to be built quickly and easily. Some of the pre-built visualizations include: the Explorer application (for GIS data browsing), Grid Alerts application and Work Orders application, which display the selected grid elements, faulty devices and required work orders, respectively, on a map.

Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Platform can access this use case by the following means:

1. This use case is implemented in TGI as part of the Explorer functionality. Consult [instructions](#) about how to access this use case in TGI. If you do not have access to a server where the use case functionality is available, contact [Awesense](#) to see a demo of it.